

State of Developer Ecosystem 2024:

Language Trends, Satisfaction, and Strategic Shifts

Key insights from JetBrains' survey of 23,262 developers on programming language usage, satisfaction, adoption intent, and domain specialization trends from 2017 to 2024.

Audience: Engineering Leaders · CTOs · Developer Experience

Teams

Source: JetBrains State of Developer Ecosystem Report 2024

Python Closes the Gap: 8 Years of Language Usage Shifts

Percentage of developers using each language (multiple selections allowed)

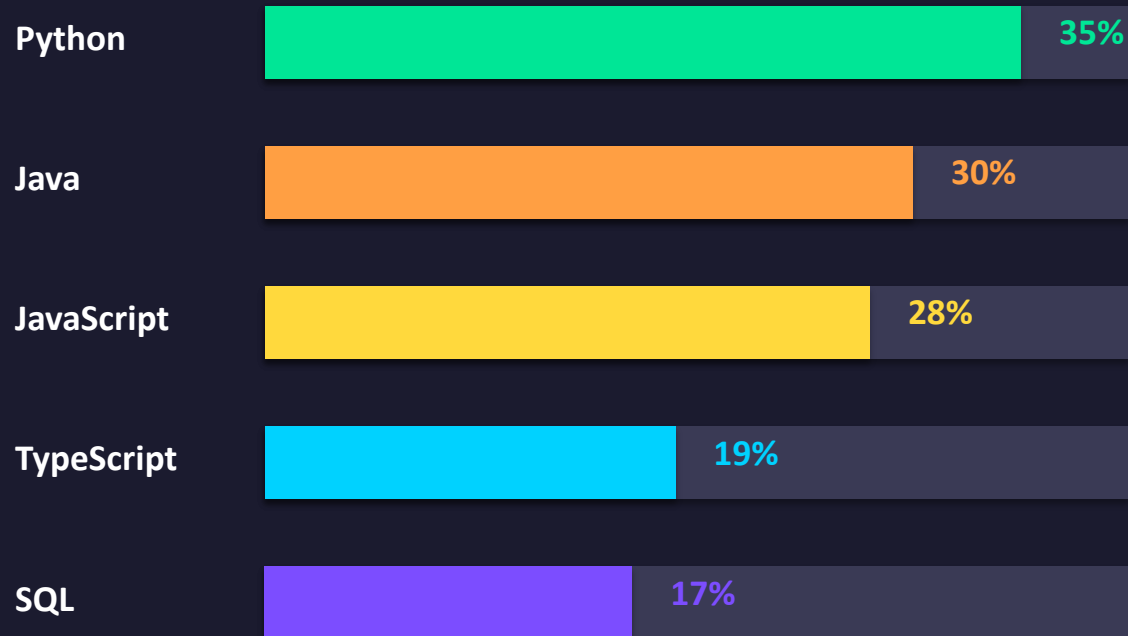
Language	2017	2020	2022	2024	Trend	
JavaScript	65%	70%	65%	61%	↓ Declining from peak	
Python	32%	55%	53%	57%	↑ Steady growth	
HTML/CSS	60%	61%	54%	51%	↓ Gradual decline	
SQL	42%	56%	49%	48%	→ Plateaued	
Java	47%	54%	48%	46%	↓ Slow decline	
TypeScript	12%	28%	34%	37%	↑ Strong growth (3×)	
Go	8%	19%	19%	18%	→ Plateaued	
Rust	—	7%	9%	11%	↑ Steady growth	
PHP	30%	27%	20%	17%	↓ Significant decline	
Kotlin	2%	17%	15%	14%	→ Plateaued	

JavaScript peaked at 70% in 2020, now 61% Python rose from 32% → 57% in 7 years TypeScript tripled: 12% → 37%

Python Is Now the #1 Primary Language at 35%

"What is your primary programming language?" vs. "Which language do you plan to adopt?"

PRIMARY LANGUAGE



ADOPTION INTENT

"Plan to adopt as primary language"

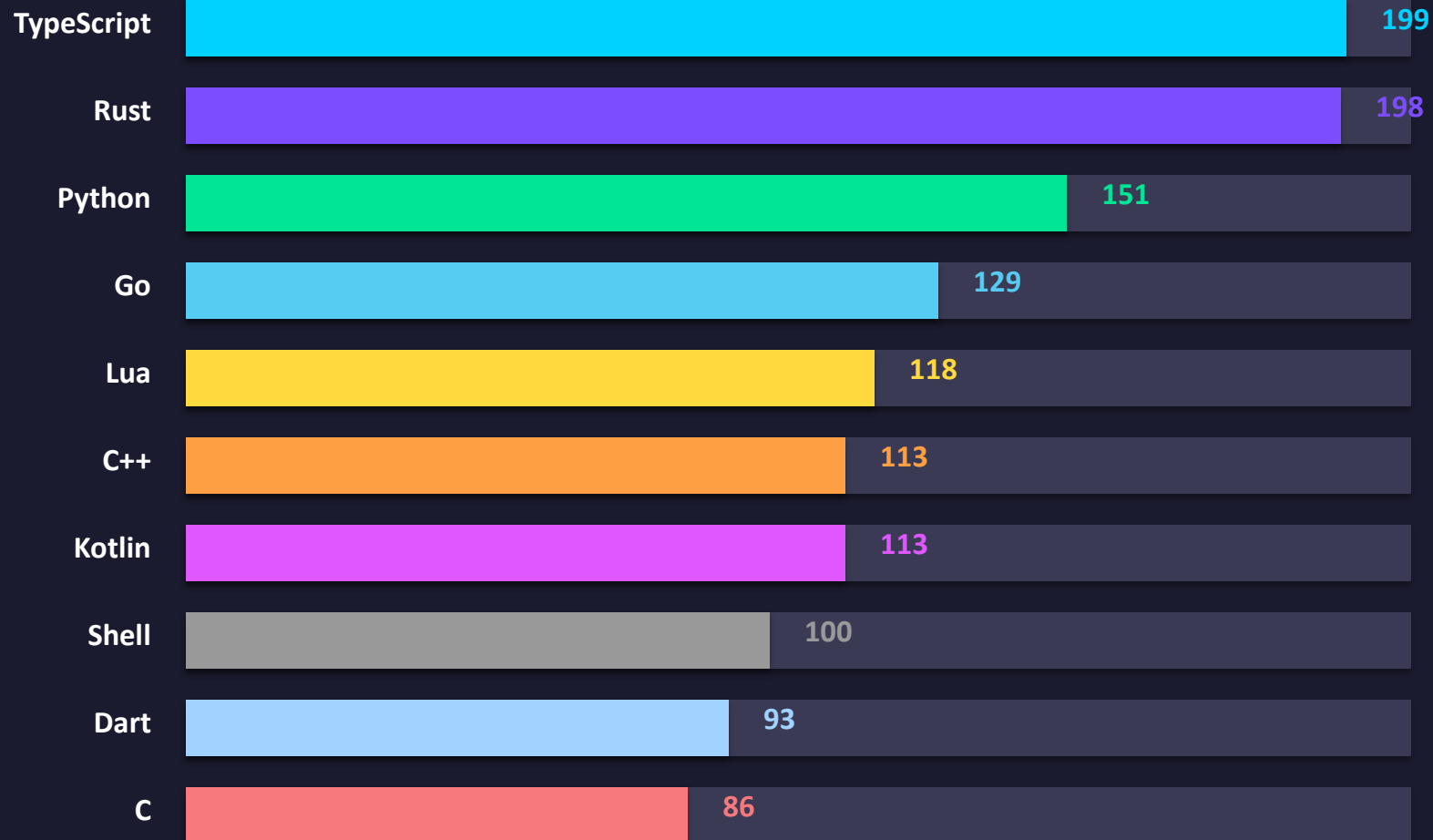


The Lock-In Effect: Developers rarely switch primary languages.

Adoption intent is below 6% for all languages — expertise investment and ecosystem maturity keep developers loyal to their chosen stacks.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Developer satisfaction score (higher = more loved)



Key Pattern

Type safety and modern tooling dominate the top of satisfaction.

TypeScript & Rust both offer strong type systems and excellent developer experience.

Satisfaction correlates with modern tooling, not raw popularity.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Top language by application domain — language choice is domain-driven, not general preference



What This Means for Your Technology Strategy in 2025

01 Invest in Python for AI/ML teams

Python is the undisputed leader in AI (31%), data science (28%), and data processing. It is the #1 primary language at 35% — the talent pool is deep and growing.

02 TypeScript is the future of frontend & full-stack

Tripled from 12% → 37% usage. #1 in developer satisfaction (199). Combined with JavaScript, it owns 58-62% of all UI development.

03 Rust & Go are the infrastructure growth plays

Go stabilized at 18%; Rust continues climbing to 11%. Both address modern needs: microservices, cloud-native, systems programming.

04 PHP & Ruby hiring pools are shrinking

PHP dropped 30% → 17%; Ruby fell 10% → 4% since 2017. Plan migration paths or accept narrowing talent markets.

05 Factor satisfaction into retention strategy

Satisfaction correlates with type safety and modern tooling. Developer experience directly impacts hiring and retention outcomes.